Maturity Wall Simulator · One-page spec

Case 02 · A Slow-Motion Crisis · Stage 04: Specification

The problem

Debt offices publish every bond they owe each month. No open tool turns that list into an answer: what rolling cheap debt into expensive debt will cost. And when an answer exists, nobody can tell where the number came from.

The promise

A forecast where every number traces back to the official file, computed bond by bond rather than approximated, that measures for itself how much the full calculation is worth.

Ten parts in four layers

Layer	Parts
Collection	1 · An adapter per debt office. 2 · Reconciliation: a gap above 0.5% of declared debt stops the load. 3 · A store of immutable snapshots, with provenance.
Computation	4 · Yield curve and scenarios, with a visible issuance mix. 5 · A monthly rollover engine that checks the debt identity at every step. 6 · Debt dynamics: when the rate overtakes growth, and which surplus stabilizes the debt.
Display	7 · Four methods on the same input, including an IMF-style approximation. 8 · The maturity wall, with assumptions attached to every chart.
Trust	9 · A reproducible permalink and an Ed25519 signature for every result. 10 · Scheduled refresh, anomaly alerts and a public changelog.

Six decisions

- Monthly steps, because short bills roll several times a year.
- A visible issuance mix, not a hidden one, because the POC showed it moves the result.
- Reconcile against declared debt instead of trusting the official file blindly.
- Measure the advantage on every run instead of assuming it exists.
- The same code on the server and in the browser, to survive load under stress.
- A permalink and a signature for every number, because the main threat is credibility.

Still open

- Against an IMF-style approximation, a gap of up to 0.53 points of GDP remained. Does the advanced version close it?
- The tool knows what each country will pay and when, but not who holds the bonds.